Medical STT Pipeline

Transcript Comparison: Predicted (Speech-to-Text) vs Ground Truth

Pipeline: Audio (.mp3) → Whisper Large-v3 (faster-whisper, GPU) → Gemini 2.0 Flash → SOAP Report

5 Representative Conversations from 5 Medical Specialties

Dataset: MTS-Dialog (Medical Transcription Scoring) | 271 audio files processed

Conversation 1 of 5

Cardiology — Chest Pain

Case ID: CAR0001

Dialogue Comparison — Cardiology — Chest Pain

First ~100 words of each transcript shown below

📄 Ground Truth Transcript

D: What brought you in today?
P: Sure, I'm I'm just having a lot of chest pain and and so I thought I should get it checked out.
D: OK, before we start, could you remind me of your gender and age?
P: Sure 39, I'm a male.
D: OK, and so when did this chest pain start?
P: It started last night, but it's becoming sharper.
D: OK, and where is this pain located?
P: It's located on the left side of my chest.
...

🎙️ Speech-to-Text Output

P: brought you in today? Sure, I'm just having a lot of chest pain and so I thought I should get it
D: checked out. Okay and before we start could you remind me of your gender and age? Sure, 39, I'm a male.
D: Okay and so when did this chest pain start? It started last night but it's becoming sharper.
D: Okay and where is this pain located? It's located on the left side of my chest. Okay and so how long
...

Summary Comparison — Cardiology — Chest Pain

SOAP summary generated from each transcript (first ~100 words)

📄 Summary from Ground Truth

The patient, a 39-year-old male, presents with 8 hours of sharp, constant left-sided chest pain that worsens with lying down and deep breaths. He reports associated symptoms of lightheadedness, dyspnea, and mild palpitations, and noted visible neck swelling without accompanying pain.

🎙️ Summary from STT Output

The patient is a 39-year-old male presenting with acute left-sided chest pain that began 8 hours ago. He describes the pain as sharp, rated 7-8/10, exacerbated by lying down and deep breathing, and associated with lightheadedness, dyspnea, and palpitations.

SOAP Report Comparison — Cardiology — Chest Pain

SOAP Field	Ground Truth	Predicted (STT)
Chief Complaint	Chest pain	chest pain
Symptoms	sharp left-sided chest pain | worsening pain on lying down | worsening pain on deep breath | lightheadedness | dyspnea | palpitations | neck...	left-sided chest pain | sharp pain | pain worsening with deep breathing | light-headedness | difficulty breathing | racing of the heart | sw...
Symptom Onset	Last night (approximately 8 hours ago)	last night (approximately 8 hours ago)
Medical History	—	—
Medications	—	—
Allergies	none	none
Exam Findings	reported visible neck swelling | patient reports 7-8/10 pain severity	neck swelling reported by patient
Diagnosis	—	—
Diff. Diagnosis	Myocardial infarction (due to family history and symptoms) | Pericarditis (suggested by positional component) | Pneumothorax | Musculoskelet...	—
Prescribed Meds	—	—
Treatment Plan	—	—
Investigations	—	—
Follow Up	—	—

Conversation 2 of 5

Musculoskeletal — Elbow Pain

Case ID: MSK0005

Dialogue Comparison — Musculoskeletal — Elbow Pain

First ~100 words of each transcript shown below

📄 Ground Truth Transcript

D: What brings you in?
P: Hi, I've I've had this pain on the outside of my uh right elbow now it's it I first started knowing, noticing it several months ago, but recently it's just been more painful.
D: OK, so you said several months ago. Um did anything happen several months ago? Was there any sort of trigger, trauma, anything like that to that area?
...

🎙️ Speech-to-Text Output

P: Hi, I've had this pain on the outside of my right elbow now. I first started noticing
P: it several months ago, but recently it's just been more painful.
D: Okay, so you said several months ago. Did anything happen several months ago? Was there
D: any sort of trigger, trauma, anything like that to that area?
P: No, there wasn't any trauma or any triggers that I noticed. I was just feeling it a bit
P: more at the end of work. Yeah, I was just feeling the pain a bit more at the end of
P: work.
...

Summary Comparison — Musculoskeletal — Elbow Pain

SOAP summary generated from each transcript (first ~100 words)

📄 Summary from Ground Truth

The patient is a healthy adult presenting with persistent right lateral elbow pain for several months, worsening with repetitive wrist movements at work. Examination findings are consistent with lateral epicondylitis, evidenced by pain on palpation of the lateral epicondyle and pain with resisted wrist extension and flexion. The patient was advised on activity modification, use of compression sleeves, continued over-the-counter analgesics, and physical therapy for recovery.

🎙️ Summary from STT Output

The patient presents with several months of worsening right lateral elbow pain, described as achy and occasionally sharp, exacerbated by wrist movement and repetitive tasks at work. Physical exam confirms pain localized to the lateral epicondyle during resisted wrist extension, with no signs of trauma or structural deformity, consistent with lateral epicondylitis.

SOAP Report Comparison — Musculoskeletal — Elbow Pain

SOAP Field	Ground Truth	Predicted (STT)
Chief Complaint	Right elbow pain	Right elbow pain
Symptoms	pain on the outside of the right elbow | pain worse with wrist bending | pain with lifting heavy objects | aching pain | sharp pain with mov...	lateral elbow pain | achy pain | sharp pain with movement
Symptom Onset	Several months ago	Several months ago
Medical History	C-section (17 years ago)	C-section 17 years ago
Medications	Tylenol | Advil	Tylenol | Advil | Ibuprofen
Allergies	No known allergies	—
Exam Findings	No redness or swelling of the elbow | No deformity or misalignment | Pain on palpation of the lateral aspect of the right elbow | No pain on...	pain with palpation of lateral aspect of elbow | pain with wrist extension (elbow in full extension) | pain with wrist flexion (elbow in ful...
Diagnosis	Lateral epicondylitis (Tennis elbow)	Lateral epicondylitis (tennis elbow)
Diff. Diagnosis	—	Fracture (ruled out due to lack of trauma)
Prescribed Meds	Continue Tylenol or NSAIDs as needed	Tylenol | NSAIDs
Treatment Plan	Activity modifications (avoid repetitive strain/extensor muscle use at work) | Use of compression sleeve | Physical therapy | Apply ice for ...	Activity modification at work (reduce repetitive wrist/forearm tasks) | Compression sleeve | Physical therapy | Apply ice as needed for pain
Investigations	—	—
Follow Up	The condition should resolve in about six months to two years at most, but normally resolves in a couple of months with physical therapy, co...	Patient was advised that the condition typically resolves in a couple of months to two years with physical therapy and conservative manageme...

Conversation 3 of 5

Respiratory — Chronic Cough

Case ID: RES0050

Dialogue Comparison — Respiratory — Chronic Cough

First ~100 words of each transcript shown below

📄 Ground Truth Transcript

D: I was wondering if you could tell me what brought you in.
P: Sure, um, I have been coughing a lot. I've been feeling short of breath and I just feel really wheezy, so I thought I should come check it out.
D: OK, when did the cough start?
P: I've always had a cough. Well, I guess I've had a cough for the past five years, but over the past three days it's gotten a lot worse.
D: OK, did anything happen 3 days ago that has brought the cough on?
...

🎙️ Speech-to-Text Output

D: I was wondering if you could tell me what brought you in?
P: Sure. I have been coughing a lot. I've been feeling short of breath, and I just feel really
P: wheezy, so I thought I should come check it out.
D: Okay. When did the cough start?
P: I've always had a cough. Well, I guess I've had a cough for the past five years,
P: but over the past three days, it's gotten a lot worse.
D: Okay. Did anything happen three days ago that has brought the cough on?
...

Summary Comparison — Respiratory — Chronic Cough

SOAP summary generated from each transcript (first ~100 words)

📄 Summary from Ground Truth

The patient, a former smoker with a history of hypertension and diabetes, presents with a chronic cough worsening over the past three days, associated with increased yellow-green sputum, wheezing, and exertional shortness of breath. The physician is investigating potential etiologies including viral infection, bacterial infection, COPD exacerbation, or COVID-19, and has ordered a PCR nasal swab and chest X-ray.

🎙️ Summary from STT Output

The patient presents with a chronic cough of 5 years duration, significantly worsened over the past 3 days with increased purulent (yellow/green) sputum and new onset wheezing. They report fatigue and increased shortness of breath with daily activities, and the physician is investigating for possible viral/bacterial infection, COPD exacerbation, or COVID-19.

SOAP Report Comparison — Respiratory — Chronic Cough

SOAP Field	Ground Truth	Predicted (STT)
Chief Complaint	Worsening cough and shortness of breath.	Worsening cough, shortness of breath, and wheezing.
Symptoms	chronic cough | worsening cough (last 3 days) | shortness of breath | wheezing | increased sputum production | yellow/green sputum | fatigue...	chronic cough | worsened cough | shortness of breath | wheezing | purulent sputum (yellow/green) | fatigue | dull chest pain (with coughing)
Symptom Onset	Chronic cough for 5 years; acute exacerbation 3 days ago.	Chronic cough for 5 years; significant worsening over the past 3 days.
Medical History	high blood pressure | diabetes	High blood pressure | Diabetes
Medications	beta blocker | metformin | calcium | vitamin D	Beta blocker | Metformin | Calcium | Vitamin D
Allergies	none	—
Exam Findings	patient reports wheezing	—
Diagnosis	—	—
Diff. Diagnosis	viral infection | bacterial infection | COPD exacerbation | COVID-19	viral infection | bacterial infection | COPD exacerbation | COVID-19
Prescribed Meds	—	—
Treatment Plan	PCR nasal swab for COVID-19 | chest X-ray | further investigations to be determined	PCR nasal swab for COVID-19 | Chest X-ray | Further investigations to determine cause of shortness of breath
Investigations	PCR nasal swab | chest X-ray	PCR nasal swab | Chest X-ray
Follow Up	Patient to follow up after investigations are completed and results are reviewed.	Patient advised to set up pneumococcal vaccination with family doctor.

Conversation 4 of 5

Gastroenterology — Abdominal Pain

Case ID: GAS0003

Dialogue Comparison — Gastroenterology — Abdominal Pain

First ~100 words of each transcript shown below

📄 Ground Truth Transcript

D: What brings you in here today?
P: Yeah hi, I'm I I have a pain in my belly. It started a couple hours ago.
D: OK, and where exactly in your belly do you have this pain?
P: It's mostly up here on the right side.
D: The right side, OK and can you describe what kind of pain is it? Is it like a dull pain or sharp pain, an aching pain?
...

🎙️ Speech-to-Text Output

D: What brings you in here today?
P: Yeah.
P: Hi.
P: I have a pain in my belly.
P: It started a couple of hours ago.
D: Okay.
D: And where exactly in your belly do you have this pain?
P: It's mostly up here on the right side.
D: On the right side?
P: Yeah.
D: Okay.
D: And can you describe what kind of pain is it?
D: Is it like a dull pain, a sharp pain, an aching pain?
P: I guess it's an aching pain, I'd say.
...

Summary Comparison — Gastroenterology — Abdominal Pain

SOAP summary generated from each transcript (first ~100 words)

📄 Summary from Ground Truth

Patient is presenting with a constant, aching right-sided abdominal pain that has persisted for 6 hours after an initial 2-hour period of intermittent pain. The patient reports a history of similar postprandial discomfort associated with fatty meals and is currently experiencing nausea but no vomiting, fevers, or bowel changes.

🎙️ Summary from STT Output

Patient presents with constant right-sided abdominal pain for the past six hours, previously intermittent and post-prandial. Patient has a history of diabetes managed with metformin and reports associated nausea but no vomiting, fever, or changes in bowel/urinary patterns.

SOAP Report Comparison — Gastroenterology — Abdominal Pain

SOAP Field	Ground Truth	Predicted (STT)
Chief Complaint	right-sided abdominal pain	right-sided abdominal pain
Symptoms	aching pain in right side of abdomen | nausea | fatigue | decreased appetite	right-sided abdominal pain | aching pain | nausea | fatigue | decreased appetite
Symptom Onset	8 hours ago	8 hours ago
Medical History	diabetes	diabetes
Medications	metformin	metformin
Allergies	—	none
Exam Findings	—	reported right-sided abdominal pain, aching quality, intensity 8/10
Diagnosis	—	—
Diff. Diagnosis	indigestion	indigestion | biliary pathology (implied by pain after fatty foods)
Prescribed Meds	—	—
Treatment Plan	—	—
Investigations	—	—
Follow Up	—	—

Conversation 5 of 5

Dermatology — Skin Rash

Case ID: DER0001

Dialogue Comparison — Dermatology — Skin Rash

First ~100 words of each transcript shown below

📄 Ground Truth Transcript

P: Hey Doc, I'm a 45 year old man coming in, I'm worried there's a little old rash on my leg that I'm getting nervous about.
D: Hi, so you I understand you have a rash on your leg.
P: I don't really know what I would call it. That's what I told the nurse when I came in, but it more hurts than anything.
D: OK, uh, how long is this been going on for?
...

🎙️ Speech-to-Text Output

P: Hey doc, I'm a 45 year old man coming in. I'm worried there's a little rash on my leg I'm getting nervous about.
D: Hi. So I understand you have a rash on your leg.
P: I don't really know what I would call it. That's what I told the nurse when I came in, but it's more hurts than anything.
D: Okay. How long has this been going on for?
P: It's been, it's been looking weird for about a week, but only, yes, it started hurting yesterday.
...

Summary Comparison — Dermatology — Skin Rash

SOAP summary generated from each transcript (first ~100 words)

📄 Summary from Ground Truth

A 45-year-old male with poorly controlled diabetes presents with a painful, swollen, and red rash on his right ankle of one week's duration, now with purulent drainage. Patient also reports constitutional symptoms including fatigue, increased hunger, and a subjective feeling of fever over the last 12 hours.

🎙️ Summary from STT Output

A 45-year-old male with a history of diabetes, high cholesterol, and chronic back pain presents with a painful, swollen, and red rash on his right ankle. The patient reports poor glycemic control (A1C ~9) and describes systemic symptoms including polyuria, hunger, fatigue, and a feeling of being hot over the last 12 hours, with purulent discharge from an open wound at the site of the rash.

SOAP Report Comparison — Dermatology — Skin Rash

SOAP Field	Ground Truth	Predicted (STT)
Chief Complaint	Painful, strange-looking rash on right ankle.	Painful rash on the right ankle
Symptoms	rash on right ankle | pain | swelling | erythema | scabbing | purulent drainage | subjective fever/feeling hot | fatigue | increased hunger …	pain | swelling | redness | polyuria | polyphagia | fatigue | feverish/feeling hot | purulent discharge from wound | dull ache | decreased s…
Symptom Onset	Rash began 1 week ago, worsening/painful since yesterday, purulent drainage started within last 12 hours.	One week (rash), one day (pain)
Medical History	Diabetes mellitus (poorly controlled, A1C ~9) | Hypercholesterolemia | History of asthma | Chronic back pain | History of sinusitis/sinus su…	diabetes | hypercholesterolemia | childhood asthma | chronic back pain | sinusitis (post-surgical)
Medications	Rosuvastatin | Metformin | Two other unspecified diabetes medications	Rosuvastatin [STT-corrected] | Metformin | Tylenol (for back pain) | Advil (for back pain)
Allergies	Penicillin (reaction: facial bloating, respiratory distress)	penicillin (facial edema, difficulty breathing)
Exam Findings	Right ankle appears swollen and erythematous | Open wound/leaking of purulent material present | Right leg feels warmer to the touch compare…	swollen right ankle | erythema | purulent discharge from open wound | warmth to touch (right leg) | left heel ulcer noted
Diagnosis	—	—
Diff. Diagnosis	Infected diabetic ulcer | Cellulitis	Cellulitis | Diabetic foot ulcer infection | Abscess
Prescribed Meds	—	—
Treatment Plan	Physical examination of the leg | Consultation with supervising physician	Physical examination of the leg | Consultation with supervising physician
Investigations	—	—
Follow Up	Patient to undergo physical examination of the leg immediately.	Patient to undergo physical exam to determine necessary next steps and consultation with the primary physician.